

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents and discusses the results obtained from the application of multiple linear regression and matrix-based numerical techniques.

4.2 Construction of the Design Matrix

The training dataset produced a design matrix of

$$800 \times 19$$

Thus,

$$X \in R^{800 \times 19}$$

The response vector contains 800 observations:

$$y \in R^{800}$$

Therefore,

$$X^T X \in R^{19 \times 19}$$

And

$$X^T y \in \mathbb{R}^{19}$$

This confirms the dimensional compatibility of the normal equations.

4.3 Estimated Regression Coefficients

Table 4.1: Estimated Regression Coefficients

Variable	Coefficient
Intercept	6.997941
Age	0.049258
Study hours per day	9.531568
Social media hours	-2.681506
Netflix hours	-2.318933
Attendance percentage	0.143439
Sleep hours	1.976635
Exercise frequency	1.319862
Mental health rating	1.953132
Gender — Male	0.139663
Gender — Other	0.807281

Pert-time job — Yes	0.344590
Diet quality — Good	-0.957974
Diet quality — Poor	-0.549902
Parental education – High School	-0.527839
Parental education – Masters	-0.701798
Internet quality –Good	-0.214797
Internet quality — Poor	-0.019558
Extracurricular participation — Yes	-0.113722

The fitted model is therefore

$$\widehat{ExamScore} = 6.9979 + 0.0493Age + 9.5316StudyHours$$

$$- 2.6815SocialMediaHours - 2.3189NetflixHours + 0.1434Attendance + 1.9766SleepHours + 1.3199ExerciseFrequency + 1.9531MentalHealthRating + \dots$$

4.4 Interpretation of Major Regression Coefficients

The coefficient of study hours per day was

9.531568

Holding other predictors constant, a one-unit increase in study hours per day is associated with an estimated increase of approximately 9.53 examination-score points.

The social-media coefficient was

-2.681506

Thus, within the fitted model, a one-unit increase in social-media hours is associated with an estimated decrease of approximately 2.68 examination-score points, conditional on the other predictors.

The Netflix coefficient was

-2.318933

Sleep hours and mental-health rating had positive coefficients of approximately

1.9766

And

1.9531

These are model-based associations and should not be interpreted as causal effects.

4.5 Model Evaluation

Table 4.2: Model Performance

Metric	Result
MAE	4.1923
MSE	26.5325
RMSE	5.1510
R^2	0.8965

The model produced

$$R^2 = 0.896531$$

Therefore, approximately 89.65% of the variation in examination scores was explained by the fitted model in the evaluated data.

4.6 Residual Analysis

Table 4.3: Summary of Residuals

Measure	Value
---------	-------

Mean residual	-0.042841
Minimum Residual	-14.764525
Maximum Residual	12.445921

The mean residual was close to zero, consistent with the least-squares property of a regression model containing an intercept.

4.7 Cross-Validation Results

Table 4.4: Five-Fold Cross-Validation

Fold	R^2
1	0.884542
2	0.913824
3	0.913516
4	0.870031
5	0.891741
Mean	0.894731
Standard deviation	0.016972

The mean cross-validation R^2 was 0.894731.

Its closeness to the primary R^2 value suggests reasonably stable predictive performance.

4.8 Comparison of Predictor Effects

Table 4.5: Standardized Coefficients

Variable	Standardized coefficient	Absolute effect
Study hours per day	14.144790	14.144790
Mental health rating	5.565088	5.565088
Social media hours	-3.129732	3.1289732
Exercise frequency	2.656671	2.656671
Netflix hours	-2.531563	2.531563
Sleep hours	2.421865	2.421865
Attendance percentage	1.349632	1.349632
Diet quality – Good	-0.463777	0.463777
Parental education – High School	-0.263867	0.263867
Parental education masters	-0.261283	0.261283
Diet quality – Poor	-0.214081	0.214081
Gender – other	0.165120	0.165120

Part-time job - Yes	0.153890	0.153890
Age	0.113344	0.113344
Internet quality – Good	-0.106590	0.106590
Gender – Male	0.069695	0.069695
Extracurricular participation – Yes	-0.053157	0.053157
Internet quality – poor	-0.007102	0.007102

Study hours per day produced the largest standardized effect.

4.9 Prediction Sensitivity

Table 4.6: Prediction Sensitivity

Variable	Prediction change
Study hours per day	10.928887
Mental health rating	4.781581
Social media hours	2.611271
Exercise frequency	2.376953
Sleep hours	2.015543
Netflix hours	1.902100
Attendance percentage	1.116296

Diet quality — Good	0.392703
Parental education — High School	0.221571
Parental education — Master	0.178615
Diet quality — Poor	0.145251
Part-time job — Yes	0.125843
Age	0.101623
Internet quality — Good	0.089734
Gender — Other	0.068401
Gender — Male	0.067821
Extracurricular participation — Yes	0.049705
Internet quality — Poor	0.005033

Again, study hours produced the largest prediction change

4.10 Sensitivity Relative to Natural Variation

Table 4.7: Numerical Variable Sensitivity

Variable	Standard deviation	Mean prediction change	Absolute sensitivity
Study hours per day	1.484922	14.153638	1.350476
Mental health rating	2.851097	5.568569	5.568569
Social media hours	1.167885	-3.131690	3.131690
Exercise frequency	2.014099	2.658333	2.658333
Netflix hours	1.092376	-2.533147	2.658333
Sleep hours	1.226013	2.423381	2.423381
Attendance percentage	9.414969	1.350476	1.350476
Age	2.302469	1.350476	0.113414

Study hours remains the most sensitive numerical predictor.

4.11 Perturbation of the Coefficient Vector

Table 4.8: Perturbation Results

Variable	Perturbation	Maximum coefficient change
Study hours per day	0.014849	0.141536
Mental health rating	0.028511	0.055686
Social media hours	0.011679	0.031317
Exercise frequency	0.020141	0.026583
Netflix hours	0.010924	0.025331
Sleep hours	0.012260	0.024234
Attendance percentage	0.094150	0.013505
Diet quality — Good	0.004844	0.004641
Parental education — High School	0.005002	0.002640
Parental education — Master	0.003725	0.002614
Diet quality- Poor	0.003896	0.002142
Gender — Other	0.002047	0.001652
Part-time job — Yes	0.004111	0.001540
Age	0.023025	0.001134

Internet quality — Good	0.004965	0.001067
Gender — Male	0.004993	0.000697
Extracurricular participation — Yes	0.004677	0.000532
Internet quality — Poor	0.003633	0.000071

Study hours remained the most sensitive numerical predictor.

4.12 Eigenvalue Analysis

The largest eigenvalue of $X^T X$ was

6,165,344,9072

while the smallest eigenvalue was

3,61694

The large difference between these values indicates substantial variation in the numerical scales represented by the matrix.

4.13 Condition Number Before Standardization

Table 4.9: Condition Numbers Before Standardization

Quantity	Condition number
X	1305.5942
$X^T X$	1,704,576.3063
$k(X^T X)^2$	1,704,576.3063

The result varies

$$k(X^T X) = k(X)^2$$

4.14 Effect of Standardization

After standardization, the largest eigenvalue was

1227.253991

and the smallest eigenvalue was

418.180781

The resulting condition numbers were:

Table 4.10: Condition Numbers Before and After Standardization

Matrix	Before	After
X	1305.5942	1.7131
$X^T X$	1,704,576.3063	2.9347

Standardization therefore produced a substantial reduction in the condition number.

4.15 Comparison of Coefficient Stability

The maximum absolute coefficient difference in one numerical comparison was

62.829434

with coefficient-difference norm

63.139991

After stabilization, the maximum difference was

0.039281

with norm

0.039852

This demonstrates the importance of numerical scaling and conditioning.

4.16 Comparison of Solution Methods

Table 4.11: Numerical Differences Between Solution Methods

Comparison	Maximum difference
Inverse vs Solve	1.3394×10^{-12}
Inverse vs Pseudoinverse	2.7169×10^{-12}
Solve vs Pseudoinverse	1.3776×10^{-12}

The extremely small differences demonstrate numerical agreement among the three approaches.

4.17 Prediction Stability Under Perturbation

The maximum prediction change was

0.002060297

The relative prediction change was

2.92946×10^{-5}

equivalent to

0.002929%

The predictions were therefore highly stable under the particular perturbation applied.

4.18 Discussion of Major Findings

The results demonstrate that matrix algebra can successfully be used to formulate and solve a multiple linear regression model.

The model produced an R^2 of approximately 0.8965, while five-fold cross-validation produced a mean R^2 of approximately 0.8947.

Study hours per day consistently produced the largest effect across the sensitivity analyses.

The numerical analysis also revealed an important distinction between predictive accuracy and numerical conditioning. The original matrix system had a relatively large condition number,

while standardization greatly reduced the condition number.

This demonstrates that a model can produce strong predictions while its underlying coefficient computation still requires numerical analysis.

4.19 Summary of Results

Table 4.12: Summary of Principal Results

Measure	Result
Total observations	1,000
Training observations	800
Testing observations	200
Model parameters	19
Design matrix	800×19
MAE	4.1923
MSE	26.5325
RMSE	5.1510
R^2	0.8965
Mean CV R^2	0.8947
CV standard deviation	0.0170

$k(X)$ before standardization	1305.5942
$k(X^T X)$ before standardization	1,704,576.3063
$k(X)$ after Standardization	1.7131
$k(X^T X)$ after standardization	2.9347
Maximum prediction change	0.0020603
Relative prediction change	0.002929%

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary of the study, conclusions drawn from the findings, contributions, recommendations, limitations and suggestions for further research.

5.2 Summary of the Study

The study investigated the application of matrix algebra to multiple linear regression for predicting students' examination scores.

A student-performance dataset obtained from Kaggle was used.

The data contained academic, behavioural, lifestyle and demographic variables.

Categorical variables were converted into dummy variables and incorporated into the design matrix.

The resulting training design matrix was

800×19

The regression model was expressed as

$$y = X\beta + \varepsilon$$

The least-squares coefficient vector was obtained through

$$X^T X \hat{\beta} = X^T y$$

The model was evaluated using MAE, MSE, RMSE and R^2 .

Additional analyses were conducted using cross-validation, sensitivity analysis, perturbation analysis, eigenvalues and condition numbers.

Standardization was also used to investigate its effect on numerical conditioning.

5.3 Summary of Major Findings

The model produced:

$$\text{MAE} = 4.1923$$

$$\text{MSE} = 26.5325$$

$$\text{RMSE} = 5.1510$$

And

$$R^2 = 0.8965$$

The mean five-fold cross-validation R^2 was

0.8947

with a standard deviation of

0.0170

Study hours per day consistently produced the largest sensitivity.

The condition number of $X^T X$ before standardization was

1,704,576,3063

After standardization, it was reduced to

2.9347

The relationship

$$k(X^T X) = k(X)^2$$

was also verified numerically.

Finally, the maximum prediction change under the final perturbation experiment was approximately

0.0020603

corresponding to a relative change of only

0.002929%

5.4 Conclusion

The study concludes that matrix algebra provides an effective mathematical framework for formulating and analysing multiple linear regression models.

The representation

$$y = X\beta + \varepsilon$$

provides a compact mathematical formulation of the relationship between the examination score and the explanatory variables.

The least-squares solution

$$X^T X \hat{\beta} = X^T y$$

provides a systematic means of estimating the regression coefficients.

The predictive results demonstrate that the fitted model performed well on the evaluated data.

More importantly, the numerical analysis demonstrated that the behaviour of the regression system depends on the conditioning of the matrices involved.

The substantial reduction in condition number following standardization demonstrates the usefulness of scaling in improving numerical conditioning.

The study therefore establishes that matrix algebra is useful not only for obtaining regression coefficients but also for investigating the numerical behaviour, sensitivity and stability of predictive models.

5.5 Contributions of the Study

The study contributes by:

1. demonstrating the practical application of matrix algebra to regression;
2. constructing and analysing a 800×19 design matrix;
3. obtaining regression coefficients using least squares;
4. verifying the regression solution through alternative computational methods;
5. investigating sensitivity and perturbation;
6. analysing eigenvalues and condition numbers;
7. demonstrating the relationship between $k(X)$ and $k(X^T X)$;

8. demonstrating the effect of standardization on conditioning;
and
9. examining prediction stability.

5.6 Recommendations

Based on the findings, the following recommendations are made.

1. Variables measured on substantially different scales should be considered for standardization when numerical conditioning is important.
2. Researchers should investigate the numerical properties of regression systems rather than relying solely on predictive accuracy.
3. Alternative numerical methods such as QR decomposition and singular value decomposition should be considered for poorly conditioned regression systems.
4. Cross-validation should be used to assess the stability of predictive performance.
5. Regression coefficients should be interpreted as model-based associations rather than automatic evidence of causation.
6. Future studies should use real-world student data where possible to determine whether the observed relationships

generalize beyond the simulated dataset

5.7 Limitations of the Study

The dataset used was simulated, limiting the generalizability of the findings.

The study also focused on multiple linear regression and therefore did not investigate nonlinear relationships in detail.

Only variables available in the dataset were considered.

Furthermore, the normal-equation approach may be affected by poor conditioning, particularly when $\mathbf{X}$ contains variables with substantially different scales.

5.8 Suggestions for Further Research

Future research could:

1. apply the methodology to real educational datasets;
2. compare linear and nonlinear regression models;
3. investigate QR decomposition and singular value decomposition;
4. examine multicollinearity more extensively;

5. investigate the effects of outliers and influential observations;
6. derive formal perturbation bounds for the regression coefficients; and
7. compare the numerical stability of different regression algorithms.

5.9 Final Conclusion

This study demonstrates that matrix algebra provides a strong mathematical foundation for multiple linear regression and predictive modelling.

The study successfully connected theoretical concepts including matrix multiplication, least squares, eigenvalues, condition numbers and perturbation analysis with a practical prediction problem.

The results show that the model achieved strong predictive performance, while the numerical analysis revealed significant improvement in conditioning after standardization.

The study therefore demonstrates the value of combining **matrix algebra, regression modelling** and **numerical analysis** in the investigation of predictive systems.

REFERENCES

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MIT OpenCourseWare. (2018). Four ways to solve least squares problems. Massachusetts Institute of Technology.

National Institute of Standards and Technology. (n.d.). Matrix condition number. NIST.

OpenStax. (n.d.). Principles of data science. Rice University.

APPENDIX A

PYTHON IMPLEMENTATION

A.1 Importing Libraries

```
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split, KFold,
cross_val_score
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
```

A.2 Loading the Dataset

```
df = pd.read_csv("student_habits_performance.csv")
```

A.3 Examining the Dataset

```
print(df.shape)
print(df.head())
print(df.info\(\))
```

A.4 Identifying Numerical Variables

```
numerical_variables = [  
    "age",  
    "study_hours_per_day",  
    "social_media_hours",  
    "netflix_hours",  
    "attendance_percentage",  
    "sleep_hours",  
    "exercise_frequency",  
    "mental_health_rating"  
]
```

A.5 Dummy Encoding

```
X = pd.get_dummies(  
    df.drop(columns=["exam_score"]),  
    drop_first=True  
)
```

A.6 Separating X and y

```
y = df["exam_score"]
```

```
X = pd.get_dummies(  
    df.drop(columns=["exam_score"]),  
    drop_first=True  
)
```

A.7 Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

A.8 Adding the Intercept

```
X_train_matrix = np.column_stack(
    [np.ones(len(X_train)), X_train.astype(float)]
)
```

```
X_test_matrix = np.column_stack(
    [np.ones(len(X_test)), X_test.astype(float)]
)
```

A.9 Computing $X^T X$

```
XtX = X_train_matrix.T @ X_train_matrix
```

A.10 Computing $X^T y$

```
Xty = X_train_matrix.T @ y_train.to_numpy()

1y)
```

A.11 Least-Squares Solution

```
beta = np.linalg.solve(XtX, Xty)
```

A.12 Predictions

```
y_pred = X_test_matrix @ beta
```

A.13 Model Evaluation

```
mae = mean_absolute_error(y_test, y_pred)
```

```
mse = mean_squared_error(y_test, y_pred)
```

```
rmse = np.sqrt(mse)
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("MAE:", mae)
```

```
print("MSE:", mse)
```

```
print("RMSE:", rmse)
```

```
print("R²:", r2)
```

A.14 Condition Number

```
condition_X = np.linalg.cond(X_train_matrix)
```

```
condition_XtX = np.linalg.cond(XtX)
```

```
print(condition_X)
print(condition_XtX)
```

A.15 Eigenvalues

```
eigenvalues = np.linalg.eigvalsh(XtX)
```

```
largest = np.max(eigenvalues)
smallest = np.min(eigenvalues)
```

```
print("Largest eigenvalue:", largest)
print("Smallest eigenvalue:", smallest)
```

A.16 Cross-Validation

```
model = LinearRegression()
```

```
kf = KFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)
```

```
scores = cross_val_score(
    model,
    X,
    y,
    cv=kf,
    scoring="r2"
)
```

```
print(scores)
print(scores.mean())
print(scores.std())
```

A.17 Standardization

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(
    X_train.astype(float)
)
```

```
condition_standardized_X = np.linalg.cond(X_scaled)
```

```
XtX_scaled = X_scaled.T @ X_scaled
```

```
condition_standardized_XtX = np.linalg.cond(XtX_scaled)
```

```
print(condition_standardized_X)
print(condition_standardized_XtX)
```

APPENDIX B

REGRESSION COEFFICIENT VECTOR

The final coefficient vector was

$\hat{\beta} =$

6.997941
0.049258
9.531568
-2.681506
-2.318933
0.143439
1.976635
1.319862
1.953132
0.139663
0.807281
0.374590
-0.957974
-0.549902
-0.527839
-0.701798

-0.214797
-0.019558
-0.113722

APPENDIX C

NUMERICAL VERIFICATION

The maximum differences obtained when comparing the three solution approaches were:

$$\textit{Inverse vs Solve} = 1.339373 \times 10^{-12}$$

$$\textit{Inverse vs Pseudoinverse} = 2.716937 \times 10^{-12}$$

And

$$\textit{Solve vs Pseudoinverse} = 1.377565 \times 10^{-12}$$

These results demonstrate numerical agreement among the three methods.

APPENDIX D

SENSITIVITY AND PERTURBATION RESULTS

The sensitivity analysis showed that study hours per day had the largest effect.

Its standardized effect was approximately

14.144790

Its prediction sensitivity was approximately

10.928887

Its perturbation produced a maximum coefficient change of approximately

0.141536

These results consistently identify study hours as the most influential predictor under the sensitivity procedures used.

APPENDIX E

EIGENVALUE AND CONDITION-NUMBER RESULTS

Before standardization:

$$\lambda_{\max}(X^T X) = 6,165,344.9072$$

$$\lambda_{\min}(X^T X) = 3.61694$$

$$k(X) = 1305.5942$$

$$k(X^T X) = 1,704,576,3063$$

After standardization:

$$\lambda_{\max}(X^T X) = 1227.253991$$

$$\lambda_{\min}(X^T X) = 418.180781$$

$$k(X) = 1.713$$

And

$$k(X^T X) = 2.9347$$

The numerical results verify:

$$k(X^T X) = k(X)^2$$

both before and after standardization.

APPENDIX F

PREDICTION-STABILITY RESULTS

The final perturbation experiment produced:

$$\textit{Maximum absolute prediction change} = 0.002060297$$

$$\textit{Mean absolute prediction change} = 0.002060297$$

$$\textit{Relative prediction change} = 2.929462 \times 10^{-5}$$

Therefore,

$$\textit{Relative prediction Change} = 0.002929\%$$

indicating very small changes in predictions under the perturbation used.